Beyond "Good Hands": Simulation Training and Objective Assessment of Medical Students' Suturing and Knot Tying Skills in the Obstetrics and Gynecology Clerkship.

Authors: Chaoul Jessica, Vann Elizabeth, Lerner Veronica, Pachipala Krithi, Ratan Rini B, Advincula Arnold, Saab Said S

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Abstract

The primary objective was to evaluate whether implementation of a structured suturing and knot-tying curriculum during the OB/GYN clerkship was associated with improvements in proficiency, as measured by a checklist-based video assessment. Secondary outcomes included changes in students' self-reported confidence in surgical knowledge and technique. Validity evidence for a checklist-based assessment of suturing and knot-tying skills was collected using Messick's framework. This was a prospective pre-post study with randomized baseline sampling of OB/GYN clerkship students. A subset completed baseline preintervention assessments, and all students underwent postintervention assessments. Both assessments utilized video-recorded checklist evaluation during a formative examination. Pre- and postclerkship surveys assessed self-reported confidence and prior surgical experience. The OB/GYN clerkship at Columbia University College of Physicians and Surgeons from December 2024 to December 2025. Participants consisted of 126 OB/GYN clerkship students. A randomly selected subset of 6 students per block (48 total) was invited to complete the preintervention skills assessment. All students completed the postintervention skills assessment at the end of the clerkship. Of 126 students enrolled, 123 were included in the analyses. Median checklist scores were significantly higher after the intervention across all skills (all $p < 0.001$). Self-reported confidence scores also improved significantly across all assessed domains ($p < 0.001$). Content and response process evidence were established through expert review of checklist items and standardized rater training. Inter-rater reliability was excellent (ICC 0.99; 95% CI 0.986-0.993). Postintervention performance was comparable to that of residents and attending physicians for most tasks. A structured suturing and knot-tying curriculum implemented during the OB/GYN clerkship was associated with significant improvements in objectively assessed technical performance across multiple foundational surgical skills.

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